J3: 3V3/GND + A0/A1; 3V3/GND + A2/A3; 3V3/GND + A4/A5

Keep VIN bypass, bootstrap and SW loop short.

33k / 10k: nominal 3.302 V. Total load and thermal check required.

Shared 3.3 V: ESP32-S3, LEDs, peripherals and camera LDO inputs.

Input: VCORE 3.3 V. Place capacitors close to the LDO.

VCORE also supplies LEDs: check total load and temperature.

Input: VCORE 3.3 V. Place capacitors close to the LDO.

VCORE also supplies LEDs: check total load and temperature.

Fit only for a verified peripheral. R31–R35: DNP in camera profile.

UART0 shared with debug; TX may emit boot logs.

GPIO3: RESET, open-drain only; JTAG strap.

GPIO46: standby; boot strap, keep low at reset.

R35: only if peripheral accepts 3.3 V control.

FPC shares I2C on GPIO4/5; pull-ups to 3.3 V.

Fixed auxiliary rails: 2.8 V / 1.5 V. Check peripheral voltage requirements.

FPC signals, fixed bias pins and voltage limits: see project documentation.

Defaults retained; optional GPIO links via R34/R35.

FPC_PULLUP: 10k to +2V8

FPC_PULLDOWN: 10k to GND

Check peripheral pinout, voltages and flex orientation.

C79/C80: +2V8 decoupling. C81: +1V5 decoupling.

AG32 universal board: ESP32 optional. See AG32 supervisor sheet and AG32-SUPERVISOR.md. Old TI/Q32 variants superseded on this branch.

U11: indirizzo I2C 0x20. Tutti i gate spenti al reset tramite pull-up P / pull-down N.

IMPLEMENTATO: solo GPIO/3V3/GND. INA, shunt, DAC e lettura ADC dedicata rimandati.

NON attivare P e N insieme. Questa revisione non aggiunge limitazione hardware.

J1/J140: PORT_A0–A5. J3: lato micro diretto. GPIO13/14: comandi isolamento.

Q3–Q8: A=P-channel; B=N-channel. U11 A–F: comandi, G: alimentazione/I2C.

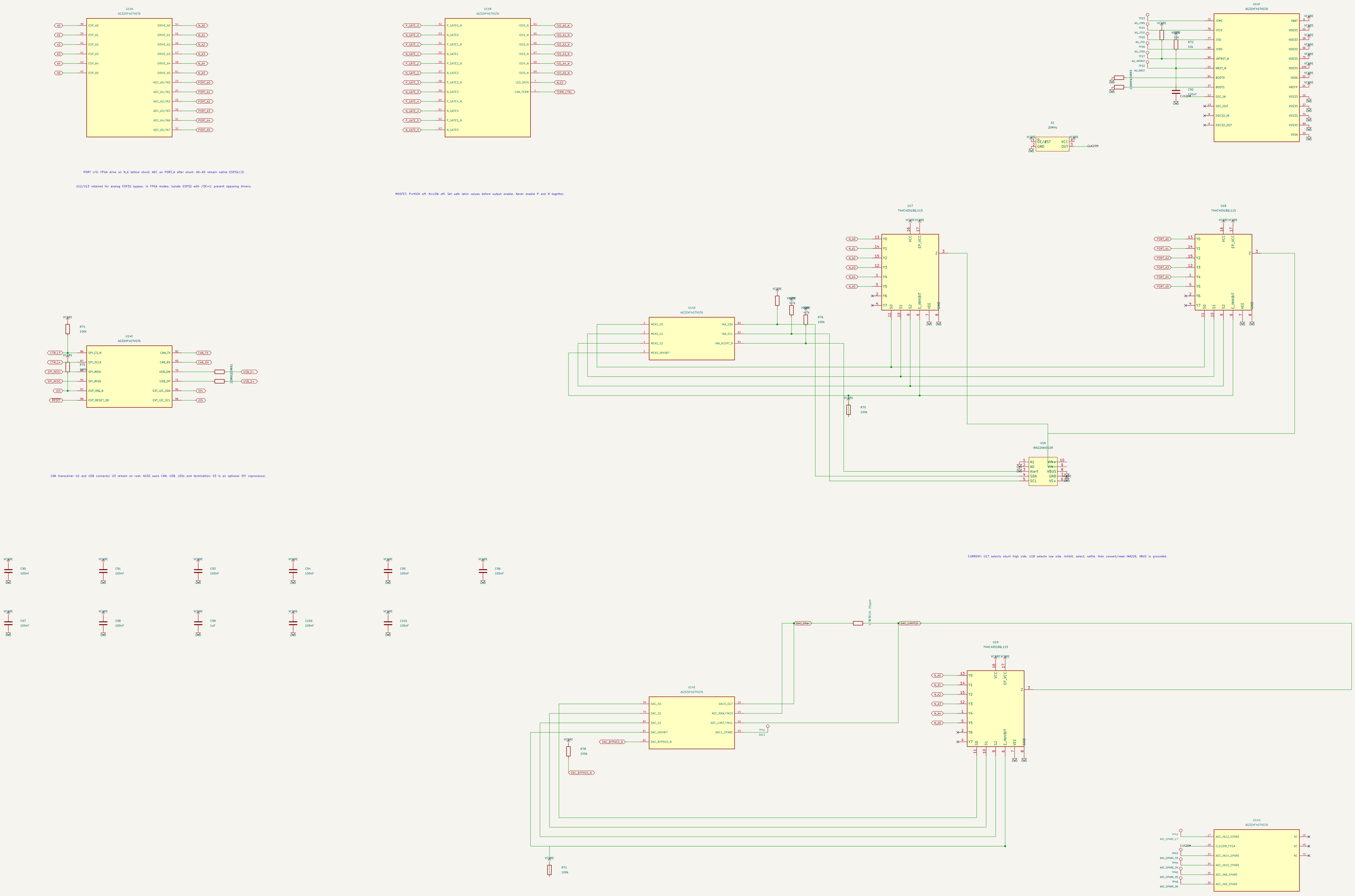
AG32 firmware: set all GPIO Hi-Z, disable both MOSFETs before switching modes. Analog bypass remains U12/U13.

A0–A5 = lato micro/J3; PORT_A0–A5 = lato MOSFET e J1/J140. Default aperto.

GPIO mode: /OE=0, P=OFF, N=OFF. 3V3/GND mode: /OE=1 prima di abilitare il MOSFET.

AG32 SUPERVISOR — ESP32 OPTIONAL / SIX UNIVERSAL PORTS

LCSC C5336269 — LQFP100. Physical pin table checked; FPGA routing/peripheral mux configuration still requires HDL/VE and bench validation.



PORT I/O: FPGA drive on N_A before shunt; ADC on PORT_A after shunt. A0–A5 remain native ESP32/J3.

U12/U13 retained for analog ESP32 bypass. In FPGA modes: isolate ESP32 with /OE=1; prevent opposing drivers.

MOSFET: P=HIGH off, N=LOW off. Set safe latch values before output enable. Never enable P and N together.

CAN transceiver U2 and USB connector U3 remain on root. AG32 owns CAN, USB, LEDs and termination; S3 is an optional SPI coprocessor.

CURRENT: U17 selects shunt high side, U18 selects low side. Inhibit, select, settle, then convert/read INA226. VBUS is grounded.

DAC: 10-bit internal DAC0. U19 routes ONE port. R77 current estimate valid only with U13C bypass open. Keep all digital drivers Hi-Z.

BOOT0/BOOT1 pulled low. VBAT, VDDA and VREFP use 3V3. No separate TI core rail remains. Clock FPGA mapping still requires VE/timing verification.

Sheet: /AG32 supervisor e strumenti/
File: ports-ag32.kicad_sch
Title:
Size: A1 | Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 2/2